



National Strategy for Mangrove Ecosystem Management

A reference for the conservation and rehabilitation of coastal areas to achieve Sustainable Development Goals and Low Carbon Development

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Executive summary

Sustainable Development Goals (SDGs) and Low Carbon Development (LCD) are global- and national-scale development agendas Indonesia is currently applying. With similar time and achievement targets where mangrove ecosystems are involved, a coordinated strategy with integrated planning is necessary.

SDGs, and particularly SDG 13 and SDG 14 (see Figure 1), are strategic goals where mangrove ecosystems can be integrated to ensure their preservation secures public attention. Though many targets have yet to be achieved, technical and non-technical challenges can be eliminated in the remaining years to come, and Indonesia can feature as a country that provides an important contribution through these essential ecosystems.

As a national agenda, LCD is closely linked to the country's *Indonesia Maju 2045* vision to become a developed nation by 2045, as proclaimed by the Government of Indonesia in 2017. Being a new paradigm, LCD still faces many challenges. However, by mainstreaming mangrove ecosystems into the country's Nationally Determined Contribution (NDC), LCD can provide a strategic impetus to 2030 and beyond through the National Long-Term Development Plan (RPJPN) to 2045. Experiences and achievements from the last two national medium-term development plans (RPJMNs) can be continued with measured and realistic targets.

The National Mangrove Ecosystem Management Strategy clearly separates between conservation goals and coastal region rehabilitation objectives as they involve extremely different ecosystem conditions, in both biophysical and social terms, as well as economic opportunities. Mangrove conservation should be prioritized bearing in mind the huge threats and challenges involved in carrying out rehabilitation and restoration, which are more technically challenging and involve greater costs.

For mangrove conservation and rehabilitation to be transparent for all stakeholders, a monitoring and evaluation, reporting, and verification system is required. This mechanism should cover all aspects of conservation and rehabilitation, including accountability over funding, and conservation and rehabilitation successes and failures.

Foreword

Indonesia's large-scale mangrove rehabilitation programme has attracted the attention of many parties both inside and outside the country. This has happened in connection with the volume of work it entails and the time available for its implementation. To ascertain public aspirations in relation to the programme, a webinar was held on 31 March 2022. The webinar discussed three main topics: a long-term coastal region management strategy; coastal regions in climate change mitigation; and coastal regions in climate change adaptation.

The expectation for this National Mangrove Ecosystem Management Strategy is that it can be made a reference for conservation and the rehabilitation/restoration of coastal regions, so that despite appearing to be different activities, they both have the same objective to achieve Sustainable Development Goals and Low Carbon Development as visions for *Indonesia Maju 2045*.

Active participation in discussions with participants triggered by presentations from speakers provided material in preparing this paper. Secondary information in the form of literature, and justification from experts affirmed the use of credible data and information to ensure this paper has its basis in scientific evidence. For that we greatly appreciate the contributions provided by various parties.

We hope this paper can become material for the preparation of the mangrove ecosystem management strategy society yearns for.

Jakarta, late July 2022

The editors

1 Introduction

1.1 Background

Mangroves are ecosystems with significant ecological, economic, and social value. At the local level, mangroves offer various ecosystem services, which include being sources of food and livelihoods for local communities. On a global level, mangroves have significant climate change mitigation potential due to their huge capacity to store carbon. With the understanding that if emissions from these large carbon stores can be avoided, then mangrove ecosystems will provide a huge contribution to achieving greenhouse gas (GHG) emissions reduction targets.

Through the Ministry of National Development Planning (PPN)/National Development Planning Agency (Bappenas), the government has set an agenda whereby wetland ecosystems, peatlands and mangroves must be integrated into national strategies. For this reason, a Strategic Coordination Team for Wetland Management for the Achievement of Sustainable Development Goals and Low Carbon Development Implementation was established through Minister of National Development Planning/Head of Bappenas Decree No. 89/2020.

In 2015, the Government of Indonesia adopted the Sustainable Development Goals (SDGs). This global agenda aims to improve community economies and well-being in a sustainable manner; ensure the sustainability of community social life; maintain environmental quality and inclusive development; and ensure governance that can guarantee an increase in the quality of life.

Then in 2017, the Government of Indonesia launched its Low Carbon Development (LCD) initiative, a development paradigm designed as part of the National Long-Term Development Plan (RPJPN) to 2045.



Photo by Mokhamad Edliadi/CIFOR

Before taking these two important decisions, the Government of Indonesia already had a regulatory framework relating to national mangrove management laid out in Presidential Regulation No. 73/2012 on a National Mangrove Ecosystem Management Strategy (SNPEM). This strategy was established in an effort to bring about sustainable mangrove ecosystem management – without disregarding community well-being – through the utilization of available resources, and constituted part of national and regional development planning. The presidential regulation also established mangrove management indicators and provided the mandate for the establishment of a national mangrove working group (KKMN) and the basis for establishing regional mangrove working groups (KKMDs).

However, through Presidential Regulation No. 82/2020 on a Committee for Handling Corona Virus Disease 2019 (Covid-19) and National Economic Recovery (PEN), several regulations were rescinded, and national committees dissolved, including Presidential Regulation No. 73/2012 and KKMN. This resulted in a vacuum in the regulatory framework relating to sustainable mangrove management. Consequently, the role of Minister of National Development Planning Decree No. 89/2020 has become important in the context of the RPJPN to 2045.

1.2 Sustainable Development Goals

Sustainable national mangrove management can support SDGs, particularly SDG 13 (Climate Action) and SDG 14 (Life Below Water). Climate change action targets to 2030 are:

- 13.1 Strengthen resilience and adaptive capacity to climate-related hazards and natural disasters in all countries.
- 13.2 Integrate climate change measures into national policies, strategies, and planning.
- 13.3 Improve education, awareness-raising, and human and institutional capacity on climate change mitigation, adaptation, impact reduction, and early warning.



Figure 1. The two Sustainable Development Goals (SDGs), SDG 13 and SDG 14, that relate to the National Strategy for Mangrove Ecosystem Management

Source: Bappenas webpage on SDGs (<https://sdgs.bappenas.go.id/tujuan-13/>)

Meanwhile, targets for the marine ecosystem sector are as follows:

- 14.1 By 2025, prevent and significantly reduce marine pollution of all kinds, in particular from land-based activities, including marine debris and nutrient pollution.
- 14.2 By 2020, sustainably manage and protect marine and coastal ecosystems to avoid significant adverse impacts, including by strengthening their resilience, and take action for their restoration in order to achieve healthy and productive oceans.
- 14.3 Minimize and address the impacts of ocean acidification, including through enhanced scientific cooperation at all levels.
- 14.4 By 2020, effectively regulate harvesting and end overfishing, illegal, unreported, and unregulated fishing and destructive fishing practices, and implement science-based management plans, in order to restore fish stocks in the shortest time feasible, at least to levels that can produce maximum sustainable yield as determined by their biological characteristics.
- 14.5 By 2020, conserve at least 10% of coastal and marine areas, consistent with national and international law and based on the best available scientific information.
- 14.6 By 2020, prohibit certain forms of fisheries subsidies which contribute to overcapacity and overfishing, eliminate subsidies that contribute to illegal, unreported, and unregulated fishing and refrain from introducing such subsidies, recognizing that appropriate and effective special and differential treatment for developing and least developed countries should be an integral part of the World Trade Organization (WTO) fisheries subsidies negotiation.
- 14.7 By 2030, increase the economic benefits to Small Island Developing States and least developed countries from the sustainable use of marine resources, including through sustainable management of fisheries, aquaculture, and tourism.

1.3 Low Carbon Development

The Government of Indonesia adopted LCD officially in 2017 under the coordination of the Ministry of National Development Planning/Bappenas. The LCD initiative aims to maintain economic and social development through low emissions development activities, minimize the exploitation of natural resources and consider the environmental carrying capacity. Meanwhile, in the achievement of LCD targets, sustainable mangrove management can curb the release of carbon emissions and contribute to economic and social sectors through the sustainable utilization and management of mangrove ecosystem services.

Achieving SDGs and LCD can contribute towards the achievement of NDC targets to reduce emissions by 31.89% unconditionally and 43.20% with international support by 2030. Therefore, a mangrove ecosystem management strategy with strong governance is needed as part of the RPJPN. This strategy must be supported by reliable and accurate data and designed to ensure that data and information are used transparently and are accessible to all stakeholders. In that way, all coordination and communication will have the same data and information reference.

2 Status of Indonesia's mangrove ecosystems

2.1 Legal status of mangroves and related institutions

National mangrove management policy references Law No. 1/2014 on Coastal Region and Small Island Management. From a conservation angle, this law not only protects mangrove ecosystems, but also regulates the cutting down of mangroves in areas already allocated for fisheries. In addition, there is Presidential Regulation No. 73/2015 on National-Level Coordination of Coastal Region and Small Island Management. However, this regulation does not specifically regulate mangrove preservation as it places more emphasis on procedures for implementing and coordinating coastal region and small island management. Meanwhile, Presidential Regulation No. 73/2012 on a National Mangrove Ecosystem Management Strategy, which had formed the main reference for mangrove management, was rescinded in 2020.

Indonesia accepted and agreed on the outcomes of the Ramsar Convention on Wetlands and laid them out in Presidential Decree No. 48/1991. As mangrove ecosystems fall under the category of wetlands, this presidential decree indirectly provides direction for the Government of Indonesia to conduct integrated mangrove ecosystem management activities in the global interest, specifically in maintaining them as habitats for waterfowl.

In addition, there are several pieces of ministerial-level legislation to consider in conducting mangrove rehabilitation:

- Minister of Forestry Decree No. 201/2004 on Criteria and Guidelines for Determining Mangrove Degradation



Photo by Donny Iqbal/CIFOR-ICRAF

- Minister of Forestry Regulation No. P.35/MENHUT-II/2010 on Amendments to Minister of Forestry Regulation No. P.32/MENHUT-II/2009 on Procedures for Preparing Technical Plans for Forest and Land Rehabilitation in Watersheds (RTKRHL-DAS)
- Minister of Maritime Affairs and Fisheries Regulation No. 24/PERMEN-KP/2016 on Procedures for Coastal Region and Small Island Rehabilitation.

To provide an institutional context, the Government of Indonesia, through Presidential Regulation No. 120/2020, established a non-structural agency tasked with accelerating the completion of mangrove rehabilitation targets called the Peatland and Mangrove Restoration Agency (BRGM). The expectation was that this presidential regulation could become a legal instrument for synchronizing mangrove rehabilitation/restoration programmes between ministries and institutions at both central and grassroots levels, and particularly institutions under the Ministry of Environment and Forestry (KLHK) and the Ministry of Maritime Affairs and Fisheries (KKP).

To support the mangrove rehabilitation policy and programme, in 2019, the Government of Indonesia prepared its National Mangrove Map. The map was prepared in stages commencing with a mangrove map of Java in 2013; Sumatra in 2014; Sulawesi in 2015; Bali and Nusa Tenggara in 2016; the Maluku Archipelago in 2017; Kalimantan in 2018; and Papua in 2019. It is worth noting that mangrove extent changed over the course of the seven years it took to prepare these maps, especially with those prepared earlier on. Consequently, the area calculation at the end of this mapping period was likely incorrect.

2.2 Ecological status of Indonesia's mangroves

According to the updated National Mangrove Map released by the Ministry of Environment and Forestry (2021), Indonesia's mangroves cover an area of 3,364,080 ha or almost a quarter of the world's mangroves (see Figure 2).

As the custodian of data on Indonesia's mangroves, KLHK conducted mangrove mapping with support from other ministries and agencies, including KKP, the Indonesian Institute of Sciences (LIPI) and the National Institute of Aeronautics and Space (LAPAN) (LIPI and LAPAN have now changed to become the National Research and Innovation Agency [BRIN]), and Geospatial Information Agency (BIG).

The National Mangrove Map 2021 (PMN 2021) is an updated version of the earlier 2019 map conducted in stages from 2013. This updated map classifies Indonesia's mangroves by canopy density and condition as dense, moderate, or sparse (see Table 1).

Indonesia has high mangrove biodiversity, with 49 species of true mangroves, or approximately 80% of the world's mangrove species (IUCN 2019). Dominant species fall under five main genera: *Avicennia*, *Rhizophora*, *Sonneratia*, *Bruguiera*, and *Nypa* (FAO 2007). The distribution of these mangrove species across different regions is influenced by several factors, including coastal typology. According to Worthington et al. (2020), there are four mangrove biophysical typologies: estuarine, deltaic, open coast, and lagoonal. Mangrove typology/geomorphology can control or regulate mangrove ecosystem carbon dynamics and also influence the choice of species and locations for mangrove rehabilitation/restoration (Sasmito et al. 2020).

Mangroves are productive wetland ecosystems, and provide various environmental, social, and economic benefits. Indonesia's mangroves have the potential to reduce national land sector emissions by as much as 30% (Murdiyarso et al. 2015), thereby playing a role in climate change mitigation and adaptation. In addition to having environmental benefits, mangroves can offer environmental services to coastal communities.

Based on data to 2019 from KKP's Coastal Ecosystem Restoration and Development Centre, mangroves play a significant role in coastal community earnings from sustainable natural resource utilization through fisheries, educational activities and ecotourism. From fisheries, fishers can obtain 5–10 kg of crabs with average daily earnings of IDR 500,000 from local sales and raw material for crab meat processing. Average daily earnings for mangrove ecotourism managers from local and foreign tourists are IDR 500,000–1,000,000 for normal days, and IDR 3,000,000–4,000,000 during holidays.

Despite mangroves being highly beneficial for communities and the environment, they are currently facing worrying rates of deforestation and degradation. Indonesia has around 476,192 ha of mangroves categorized as being in critical condition inside the forest estate and 161,432 ha outside the forest estate. Latest research indicates that oil palm plantations and aquaculture are the main causes of mangrove loss in Indonesia (Richards and Friess 2016).

Based on data from land cover maps, mangrove cover in Indonesia fell by 567,845 ha from 1990 to 2019 (see Table 2). Meanwhile, 32,662 ha of mangroves outside the forest estate have been declared critical and beyond rehabilitation, and 6,305 ha critical with rehabilitation remaining viable.

In striving to reduce mangrove degradation rates, the Government of Indonesia has undertaken various conservation and rehabilitation/restoration programmes. One such effort was its establishment of the Peatland and Mangrove Restoration Agency (BRGM), which was formalized in December 2020 through Presidential Regulation No. 120/2020. BRGM has a presidential mandate to rehabilitate 600,000 ha of mangroves in nine provinces (North Sumatra, Riau, Riau Archipelago, Bangka Belitung, West Kalimantan, North Kalimantan, East Kalimantan, West Papua, and Papua) by 2024.



Figure 2. National Mangrove Map, Ministry of Environment and Forestry (2021)

Table 1. Classification of mangroves in Indonesia based on canopy density and condition

Canopy density	Class	Area (ha)
Dense	Intact	3,121,240
Moderate	Intact	188,366
Sparse	Degraded	54,474

Table 2. Mangrove cover in Indonesia

Mangrove type	Area (ha)	
	1990	2019
Primary	1,859,735	1,522,422
Secondary	1,677,374	1,446,842

Source: Ministry of Environment and Forestry (2019) land cover maps

Further, in reducing mangrove degradation rates in Indonesia, KLHK and KKP have undertaken rehabilitation/restoration efforts for critical mangrove ecosystems – where such efforts are still viable through mangrove planting – in 18 regencies/municipalities and established mangrove nurseries in 12 regencies/municipalities. In addition to these efforts, the two ministries are also raising community awareness on the importance of mangrove ecosystems for life.

Another important thing for the government and other stakeholders, including regional governments, the private sector, and local communities around mangrove areas, to do is maintain intact mangroves and ensure they are utilized in a sustainable manner.

3 Opportunities and challenges in mainstreaming mangrove ecosystems

3.1 The run up to 2030

Wetland ecosystems, including mangroves, have a special place in global agendas such as the SDGs and the Nationally Determined Contributions (NDCs) laid out in the Paris Agreement. This is because coastal and marine ecosystems have been the subject of global deliberations in events including the Our Ocean Conference, Ramsar Convention on Wetlands, UN Framework Convention on Climate Change, UN Convention on Biological Diversity, and UN Decade on Ecosystem Restoration. Coastal and marine issues have even entered the intensive dialogue stage towards their integration into the public agenda.

For Indonesia, the decade from 2020 to 2030 is a period filled with challenges and opportunities, particularly in mainstreaming the roles of its extensive carbon-rich mangrove ecosystems. Several programmes and projects are being developed to mainstream mangroves as a nature-based solution. However, as this topic and approach remains new, it still poses various challenges alongside the many opportunities it presents.

Table 3 shows various opportunities and challenges in mainstreaming mangroves on a global scale through SDGs, and a national scale through LCD in the less than one decade still remaining. These opportunities generally relate to Indonesia's position as a developing country that is proactive towards global agenda dynamics, and particularly climate change.

The hard work of stakeholders facilitated by the government at various negotiation tables in both maritime law and climate change conventions has borne fruit, as has its implementation, as gauged by transparent monitoring, reporting, and verification (MRV) systems.



Photo by Kate Evans/CIFOR

Table 3. Opportunities and challenges in mainstreaming mangrove ecosystems into Sustainable Development Goals (SDGs) and Low Carbon Development (LCD)

	Opportunities	Challenges
SDGs	Archipelagic outlook and maritime law (UNCLOS)	Overfishing, illegal fishing, and destructive fishing
	Food and livelihood source	Coastal area degradation
	Length of coastline	Plastic waste and marine pollution
	Fisheries subsidies through the WTO	Impacts of marine acidification
	Market access for local fishers	Utilization of marine science and technologies
LCD	Calculation methodologies available	Mangrove deforestation
	Adaptive and dynamic NDC	Low planting success rates
	Forestry and other land uses (FOLU) net sink	Mangrove species biodiversity
	Carbon tax and economic value	Poverty in coastal communities
	Community and local government participation in coastal regions	Stakeholder coordination and regulatory framework
	Employment opportunities in land sectors	Education in coastal communities

Meanwhile, the list of challenges in Table 3, which represents various SDG targets not achieved by 2020, remains long. Overall, the achievement of these targets was probably constrained by the Covid-19 pandemic. Nevertheless, their achievement in the remaining time remains a challenge.

Challenges that will be faced in using mangrove ecosystems in LCD implementation can be categorized as technical, such as determining appropriate rehabilitation locations, and non-technical linked to community acceptance of rehabilitation activities. Rehabilitation programmes to reduce carbon emissions will encounter difficulties in achieving targets in the near future because large volumes of carbon are in protected rather than degraded and rehabilitated mangroves.

In other words, LCD using mangrove ecosystems should involve large-scale efforts to protect conservation areas. Only in this way can a repeat of the huge emissions from mangrove deforestation over the past few decades be avoided. This scenario must become a strategic national programme.

3.2 Towards *Indonesia Maju 2045*

Sustainable natural resources management is the key to success for economic development in Indonesia's vision to become a developed nation by 2045. *Indonesia Maju 2045* contains several pillars for the country's development, one of which is the Pillar of Sustainable Economic Development. The potential in wetlands, including mangrove ecosystems, can support a number of aspects in this pillar.

To help Indonesia achieve its vision for 2045, a long-term National Strategy for Mangrove Ecosystem Management needs to consider four major components:

- **Increasing mangrove extent:** Rehabilitation should have a special focus on open, disaster-prone areas, particularly those most at threat from rising sea levels. Mangroves are *green infrastructure* that can be a nature-based solution to handling the impacts of climate change. This increase in extent is the net area after rehabilitation compared with that before.
- **Guaranteeing biodiversity:** Ensure that floral and faunal biodiversity in mangrove ecosystems is restored and safeguarded. All forms of rehabilitation that separate species biodiversity and mangrove habitats must be avoided. Large-scale planting has the potential to result in monocultures that will degrade natural ecosystems.

- **Emissions reductions:** With their large carbon reserves, mangrove ecosystems will be effective in mitigating atmospheric emissions, and further emissions can be avoided through the conservation of intact mangroves, which store large volumes of carbon in the soil or substrate underneath them. Emissions reductions should be calculated for various mangrove categories and recorded as part of the land sector. A 'stock-difference' approach should be applied in accordance with Intergovernmental Panel on Climate Change (IPCC) recommendations, and calculated for all components (aboveground biomass, belowground biomass, necromass, and soil).
- **Economic gains:** As people in coastal areas expect employment opportunities and livelihood sources, any forms of rehabilitation or restoration must think about long-term work opportunities for the local populace. Calculating numbers of work-hours in regard to involving locals in a rehabilitation/restoration project can provide an erroneous signal on community livelihoods in coastal regions.

Indonesia Maju 2045 is marked by improvements in environmental quality, including in coastal and marine areas, and reductions in absolute poverty, particularly in the poorest communities in coastal areas. Means for achieving these should be designed for laying out in five national medium-term development plans (RPJMNs). The current government led by the president with a mandate to 2024, could begin this process now.

In the approach to 2045, stages in mangrove rehabilitation/restoration should be planned, implemented, evaluated and 'released' as programmes that can be audited thoroughly in relation to their survival rates for planted mangroves. Table 4 illustrates stages in this process in the national long-term development plan (RPJPN).

Table 4. Five stages in mangrove rehabilitation to achieve LCD targets to 2045 in five national medium term plans or one long term development plan

	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
National Medium-Term Development Plan (RPJMN) 2022–2024	1				
RPJMN 2025–2029		2 1			
RPJMN 2030–2034			3 1,2 1		
RPJMN 2035–2039				4 1,2,3 1,2	
RPJMN 2040–2044					5 1,2,3,4 1,2,3

Note:

	Target (ha)		Evaluation (% survival)		Release (% success)
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Figures in the column denote the order or stage in determining targets in one RPJMN.

Bearing in mind the length of this process, it is necessary to anticipate changes in government, and the possibility of different development agendas and priorities. To guarantee sustainability and accountability in mangrove rehabilitation/restoration programmes, a transparent MRV system is required. This system should be protected by a definitive and strong legal framework and governance so public funds can be accounted for in the long term.

This is to ensure that in mangrove ecosystem management, *Visi Indonesia 2045* is not reviewed only from an environmental perspective, but also from social and economic perspectives. Sustainable natural resources management is key to success for economic development as laid out in *Visi Indonesia 2045*. As a new paradigm, the LCD concept should be implemented and evaluated so that through the land sector, Indonesian communities, and particularly those domiciled in coastal regions, have strong climate resilience.

4 Mangrove ecosystem management strategies

4.1 Mangrove ecosystem conservation

Frequently, efforts relating to mangrove ecosystems are limited only to rehabilitation/restoration. However, conservation efforts to preserve intact mangrove ecosystems also play a vital role for the continued existence of mangrove ecosystems in Indonesia. Consequently, conservation is an essential part of any mangrove ecosystem management strategy.

Categories of the 3.31 million ha of mangroves according to the National Mangrove Map (Ministry of Environment and Forestry 2019) are detailed in Table 5. Despite the highly general nature of these categories, the table provides an illustration that a large proportion of mangroves (81%) remains intact. This means efforts to protect this large extent of mangroves will require significant resources.

This categorization, showing areas in terms of jurisdiction or legal control by KLHK, is not helpful in terms of determining the overall sustainability of mangroves. These areas are highly likely to border areas where, if problems such as conflicts arise, then any resolution will be extremely long-winded, or cases may even be neglected and remain unresolved for a very long time.

From this categorization, 637,624 ha of mangroves, or 19% of the total mangrove area both inside and outside the forest estate, have been declared degraded and in need of rehabilitation by BRGM. The majority of these critical mangroves (487,988 ha) are in nine provinces prioritized for rehabilitation, while the remainder are in 25 other provinces, including densely populated ones on Java.



Photo by CIFOR-ICRAF

Table 5. Mangrove distribution by criticality

	Mangroves inside the state forest lands (ha)	Mangroves outside the state forest lands (ha)	Total
Non-critical	2,057,295	616,287	2,673,583 (81%)
Critical	460,244	177,380	637,624 (19%)
Total	2,533,488	777,719	3,311,207

Source: Ministry of Environment and Forestry (2020)

Table 6. Distribution and area of mangroves by forest estate category in the updated National Mangrove Map

Category	'Existing' mangroves (ha)	'Potential' mangrove habitat (ha)	Total (ha)
Protection forest	911,397	83,737	995,134
Conservation forest	748,271	48,837	797,108
Production forest	1,001,614	142,961	1,144,575
Other use area (APL)	701,798	480,648	1,183,446
Total	3,364,080	756,183	4,120,263

Source: Ministry of Environment and Forestry (2021)

Two years after being published and used to localize rehabilitation/restoration areas, the National Mangrove Map (PMN) was revised and updated with the categories shown in Table 6. The updated map uses the terms 'existing' and 'potential' mangroves, which when totalled, cover a total area of 4.12 million ha. It is unclear whether 'potential' mangrove habitats are identical to or overlap the critical mangroves in the earlier version of the map. Nevertheless, it is certain that by function, areas other than other use areas (APL) are inside the forest estate.

Table 6 also shows the majority of mangroves (3.36 million ha) constituting 'existing' mangroves. This largest proportion is easier to protect and should be protected. The government should not be negligent in securing these assets, including the carbon and environmental services therein.

Important things that both the government and other stakeholders should consider and immediately implement in any mangrove conservation programme are:

- strengthening the mangrove ecosystem conservation regulatory framework, policies, and institutions
- strengthening data and information management and using the latest science and technologies to support mangrove ecosystem conservation
- enhancing the roles and capacities of communities as mangrove ecosystem conservation partners
- enforcing the law in mangrove ecosystem conservation
- encouraging funding mechanisms that support conservation as a profit centre rather than a cost centre

An important thing to do in strengthening the regulatory framework, policies, and institutions in mangrove ecosystem management, and particularly for the conservation aspect, is to provide enabling conditions. Therefore, this step should be taken not only at the national and provincial levels, but at the grassroots level as well so government initiatives can directly impact issues and challenges being faced.

If these institutions are formed and operational, it is important to ensure the regulatory frameworks under their remit can be harmonized and synchronized. Capacity building for conservation institutions should include a regulatory awareness agenda that can strengthen and reaffirm law enforcement, including the imposition of sanctions on violators of mangrove conservation policies.

Strategies surrounding spatial utilization should be integrated into all levels so that overlapping land uses and legal uncertainty can be minimized or even eliminated. Integrated steps in spatial planning will strengthen the legal status of mangrove conservation areas.

Support with reliable data and information, and a capacity to make the latest information available and accessible to a variety of stakeholders are essential requirements. Reliable data using precise methods can become an information source and reference in implementing mangrove ecosystem protection.

Science and technology, particularly new knowledge generated in scientific circles, should be made a support in mangrove conservation decision making on the basis that it is scientifically proven and has been peer reviewed in an expansive network. All forms of resources and facilitation should be explored to improve the quality of scientific and technological output relating to mangrove management and conservation.

Mangrove ecosystem conservation will be applied more effectively with an awareness from all stakeholders, including communities, of the importance of mangrove ecosystems for nature and humankind. Increasing communities' active roles in conservation and sustainable utilization is equally important, because essentially, communities are mangrove conservation partners.

Clarity over law enforcement systems will also play an important role in conservation efforts for natural resources, including mangrove ecosystems. Strengthening law enforcement in the implementation of mangrove ecosystem rehabilitation, protection, and management is an important policy direction for achieving sustainable mangrove ecosystem management targets in Indonesia.

All things that need to be done and paid attention to in mangrove ecosystem conservation efforts should be supported by a sustainable funding mechanism that is not a burden on public funds. Therefore, it is important to be able to collaborate by involving various parties to prepare a funding scenario, especially for raising funds outside national APBN and regional APBD budgets. This funding mechanism should accord with provisions in prevailing legislation, so it does not give rise to difficulties in audits or financial inspections at a later date.

Carbon trading and taxes are new schemes that should be integrated into mangrove ecosystem conservation in the context of avoided emissions, as conservation principles are applied with good governance.

4.2 Mangrove ecosystem rehabilitation

The rehabilitation of approximately 600,000 ha of degraded mangrove ecosystems, or 19% of all Indonesia's mangroves (see Table 5), will require different techniques to those used for terrestrial forests. These unique habitats, marked by the mixing of fresh and sea water with mud or sand substrates, constitute a challenge in their own right in replanting mangrove seedlings. Species selected for planting must be appropriate for the habitat, and large-scale planting of monocultures must be avoided.

Mangroves often rehabilitate themselves naturally, and humans can follow these processes. Formations of natural mangrove stands should be reproduced and be made – strategies that can be replicated in other places with similar habitat conditions.

Engaging communities impacted directly by mangrove degradation could be an option in determining an adaptive rehabilitation model. Involving a community agenda could help make a rehabilitation programme a success as it would benefit the community. In that way, mangrove rehabilitation would become a framework for improving and maintaining a socioecological system in a sustainable manner (Kibler et al. 2018; Krievins et al. 2018).

Some strategies applied in conservation could be adopted or modified to conditions on the ground with the main aim of restoring habitats and ecosystems from degradation that may have gone on for years. However, rehabilitation/restoration efforts should pay specific attention to the following:

- synchronizing of regulatory frameworks from various sectors and levels associated with mangrove rehabilitation
- improving the availability of and access to high quality data and information for all stakeholders in mangrove rehabilitation
- strengthening institutions and governance to ensure monitoring, evaluation and reporting for verifying mangrove rehabilitation implementation
- building awareness, capacity and participation of communities as partners in mangrove rehabilitation/restoration
- developing blended funding schemes from sources other than the national APBN budget

Past failures were frequently linked to a lack of coordination between government institutions, leading to overlapping rehabilitation programme implementation. This resulted in low success rates and significant funds being wasted, with outcomes that were difficult to account for. It must be easier to trace funds to gauge their success.

In an effort to rectify past mangrove rehabilitation failures, the government has undertaken several corrective actions, such as KLHK and BRGM preparing the National Mangrove Rehabilitation Roadmap in 2022 as a guide and reference for stakeholders in preparing their mangrove ecosystem management action plans. In addition, the Government of Indonesia has also established the Indonesia Environment Fund (IEF) to become a financing mechanism for environmental protection and management. One of IEF's roles is to accumulate and channel funds for the environment, including those relating to mangrove ecosystem protection and management, with the expectation that the funding mechanism can be more easily monitored and more accountable.

Data and information (including financial information) must also be easily accessible to all stakeholders involved in rehabilitation so programme planning and implementation can be more synchronous. With good governance and simple bureaucracy, important rehabilitation partners, namely communities, will become involved in mangrove rehabilitation. The alignment of perceptions in ensuring the achievement of multistakeholder objectives will streamline rehabilitation programme implementation and administration.

Per hectare costs for mangrove rehabilitation can reach USD 3,900 (World Bank 2022). It will be difficult to raise sufficient sums by relying only on government-managed public funds (APBN). Sustainable and successful mangrove rehabilitation also requires blended funding and non-budgetary funds from private businesses and philanthropy. Overseas trust funds (bilateral and multilateral) commonly associated with global climate change and biodiversity schemes should be explored in connection to national interests and agendas. In addition, rehabilitation funding scenarios and mechanisms must remain transparent to increase public trust in their accountability.

4.3 Benefit-cost ratios for mangrove conservation and rehabilitation

A benefit-cost ratio (BCR) is a financial instrument that can be used for determining choices. Their values can be calculated from the financial benefits the goods and services that mangrove ecosystems provide, and the costs required to maintain them so the goods and services produced can remain sustainable.

Mangrove conservation and rehabilitation will have different components; some beneficial and others burdensome. Differences will also occur from one location to another in relation to infrastructure, market access, and the economic value of goods and services produced.

Services provided by mangroves include:

- Regulating services: climate change, inundation, erosion, pollution
- Provisioning services: food, fuel/energy
- Supporting services: nutrient cycle, sedimentation
- Culture: natural beauty, recreation, education, festivals/spiritual events

Goods produced, though often relating to environmental services provided by intact mangroves, can also take the form of goods that when exploited can result in permanent damage to mangrove ecosystems (mining/excavation materials, wood, and charcoal).

BCR analyses conducted by the World Bank (2022) in various coastal regions in Indonesia generally indicate higher BCR scores for mangrove conservation (2–5) than for mangrove rehabilitation (1–2). Therefore, as its BCR is noticeably higher across the country, conservation should be the main priority. Nevertheless, rehabilitation is also important, and should be integrated with conservation activities to ensure it has a higher chance of success and lower risk of failure.

5 Monitoring, reporting, and verification

5.1 Monitoring and evaluation

Evaluation of the monitoring, reporting, and verification (MRV) of programme implementation is an integral part of developing any mangrove ecosystem management strategy. MRV is carried out to obtain information on progress made in mangrove ecosystem management, and lessons relating to things requiring correcting and improving in mangrove ecosystem management governance. Coordination and consolidation between stakeholders can also be improved through information obtained from MRV processes.

Based on Law No. 25/2004 on the National Development Planning System, and Government Regulation No. 39/2006 on Procedures for Control and Evaluation of Development Plan Implementation, there are four development planning stages:

- planning preparation
- planning establishment
- planning implementation control
- planning implementation evaluation

The stages involved in monitoring government programmes are laid out in Government Regulation No. 39/2006. The connotation for control in this regulation is as part of a series of management activities intended to ensure a programme or activity is implemented in accordance with a set plan. Development planning implementation control should be documented in the form of reports so any data and information gathered can be accessed by other parties that may need them.



Photo by Aulia Erlangga/CIFOR-ICRAF

Monitoring and evaluation of mangrove ecosystem management should be carried out through several stages and on all aspects. These stages should describe input, process, output, and outcome indicators. Monitoring conducted at every stage should aim to create adaptive management, which constitutes a process that has a continual structure and systematics; and corrects decisions, management policy, and implementation by studying the impacts and consequences of earlier decisions. This adaptive management can occur during every stage of the process.

Details of monitoring and evaluation activities that should be carried out are as follows:

- Regional monitoring and evaluation teams should be established by regional head decree, and a central monitoring and evaluation team should comprise related ministries and agencies at the central level.
- Regional monitoring and evaluation teams should be made up of representatives of regional government offices associated with peatland and mangrove ecosystem management in their respective regions.
- The central and regional monitoring and evaluation teams should report their activities using the National Low Carbon Action Plan Planning and Monitoring Application (AKSARA), which is currently undergoing development.

To facilitate monitoring and evaluation processes, monitoring and evaluation of the National Wetlands Management Strategy will be integrated into the Low Carbon Development monitoring and evaluation system using the Bappenas-developed AKSARA application.

5.2 Reporting

Reporting and evaluation should incorporate monitored criteria or variables and use indicators of mangrove ecosystem management performance. Details of monitored variables should be assigned clear confines and definitions. On this basis, reporting should be integrated into national and regional long- and medium-term development plans, and the strategic plans (*Renstra*) of every related ministry/ agency or regional government.

Reporting of mangrove ecosystem management monitoring and evaluation activities should take place every year. Bappenas will undertake this verification and reporting, and regional development planning agencies (Bappeda) should validate reports.

5.3 Verification

Verification and validation of reports should be coordinated with the finance sector in regard to use of funds. Despite not constituting financial audits conducted by relevant authorities, they remain necessary in the context of multisectoral and multilocal management coordination.

Verification and validation should follow RPJMN performance monitoring and evaluation schedules, which take place twice in one period: at the middle and at the end of each one.

6 Closing

As mangroves are unique ecosystems, their management requires special handling from both technical and institutional angles. These ecosystems in transitional zones between terrestrial and marine ecosystems necessitate specific requirements for their sustainability. These ecological locations provide their own challenges when it comes to coordinating programmes and activities that relate to the sectoral authority of different government ministries and agencies.

The time scales necessary for ecosystem management – which is highly vulnerable to both natural and anthropogenic disturbances – require long-term conservation strategies and rehabilitation/restoration stages.

Conservation of existing mangroves should be prioritized, bearing in mind rehabilitation success rates remain low. Meanwhile, any rehabilitation of degraded mangroves or areas with potential for ecosystem restoration requires careful and thorough planning to increase the likelihood of success.

Legislatively protected MRV is a reliable tool that must be developed and implemented immediately after a strategy is executed. The costs and development of an MRV apparatus should be included in any strategy development process.



Photo by Sigit Deni Sasmito/CIFOR

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Sustainable Development Goals and Low Carbon Development are global- and national-scale development agendas Indonesia is currently applying. Mangrove ecosystems are included to achieve the targets within the same time frame, and a coordinated strategy and integrated planning are necessary

The National Strategy for Mangrove Ecosystem Management clearly separates between conservation and rehabilitation goals of coastal region rehabilitation objectives as they involve extremely different ecosystem conditions, in both biophysical and social terms, as well as economic opportunities. Mangrove conservation should be prioritized, bearing in mind the huge threats and the challenges involved in conducting rehabilitation and restoration, which are more technically challenging and involve greater costs.

For mangrove conservation and rehabilitation to be transparent for all stakeholders, a monitoring and evaluation, reporting, and verification system is required. This mechanism should cover all aspects of conservation and rehabilitation, including accountability over funding, and the performance (successes and failures) of conservation and rehabilitation efforts.



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